



The influence of environmental factors and phytoplankton blooms on cyphonautes larvae abundance and bryozoan colony development at a kelp farm in central Norway

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ABSTRACT

Seaweed cultivation offers a potentially sustainable solution for biomass production. However, in Norway, biomass quality at kelp farms is affected by biofouling, typically from encrusting bryozoans, such as *Membranipora membranacea* and *Electra pilosa*. This study investigated the drivers of bryozoan biofouling at a kelp farm in the coast of central Norway in 2022 and 2023. Environmental variables (temperature, salinity, turbidity, light, nutrients and wind), phytoplankton concentrations (chlorophyll *a* and size structure), and bryozoan (cyphonautes) larval size, abundance and recruitment on the kelp species, *Saccharina latissima*, were monitored. Phytoplankton biomass and size structure were monitored because cyphonautes are planktotrophic, therefore, phytoplankton was used as a proxy for food availability. Spring phytoplankton blooms (up to ~ 6 mg chlorophyll *a* m⁻³) followed increased irradiance and reduced mixing, with cyphonautes larvae showing two main abundance peaks – in April (~200–400 ind m⁻³), 1–2 weeks after the onset of the bloom – and in June (~450 ind m⁻³). Larval abundance was associated with low salinity (value ~ 32), stratified, fresher coastal waters. *Membranipora membranacea* larvae were generally more abundant and reached larger sizes (up to 0.6 mm in length) during the spring settlement period (late April–June). Larval size, rather than abundance alone, was most closely related to subsequent colony settlement, highlighting the importance of larval maturity for predicting biofouling risk. Colony abundance, size, and areal coverage were higher in earlier-deployed kelp (October 2022 versus January 2023) and increased exponentially from May (<1 %) to late June (up to 11 %). Recruitment peaked during a sharp increase in temperature (1–2 °C in a few days) and was low during a period with high wind speed (up to 15 m s⁻¹). Our findings demonstrate that food availability, water column stability, rapid temperature increases and cyphonautes size structure are the dominant factors influencing bryozoan biofouling on kelp.

1. Introduction

There is a need for sustainably sourced human food and animal feed (FAO, 2022). Cultivated seaweed can be a sustainable biomass which contributes to food security, due to its high nutritional value and low environmental impact and CO₂ footprint (Imchen, 2021; Koesling et al., 2021; Thomas et al., 2021). While the industry is well-established in Asia, seaweed cultivation at sea has been gaining momentum in Europe

over the last 10 years, particularly for the brown macroalgal kelp species *Saccharina latissima* ((Linnaeus) C.E.Lane, C. Mayes, Druehl & G.W. Saunders et al., 2010) and *Alaria esculenta* ((Linnaeus) Graville 1830; Afonso et al., 2021; Forbord et al., 2020). However, the industry is still very much in its infancy compared to Asia, and seaweed cultivation in Europe and Norway is still not profitable at the current scale (Moscicki et al., 2025).

Biofouling by the encrusting colonies of bryozoan species

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Membranipora membranacea (Linnaeus, 1767) and *Electra pilosa* (Linnaeus, 1761) is one of the main factors that limits high quality yield of sea cultivated kelp (Skjermo et al., 2014). The rapid appearance of bryozoan colonies on the kelp fronds (days to weeks in spring/summer) damages the kelp lamina and reduces its commercial value, forcing the seaweed farmers to cut short their harvest when kelp growth is at its peak (Førde et al., 2016). In Norway, settlement and colonial growth of *M. membranacea* and *E. pilosa* on cultivated kelp vary according to latitudinal gradient, following the rise of temperature and availability of food (phytoplankton blooms) (Forbord et al., 2020; Førde et al., 2016). Wave exposure has also been shown to strongly influence bryozoan settlement and biofouling dynamics (Visch et al., 2020), and wind speed can be used as a proxy for wave exposure when assessing its influence on larval settlement and colony development. However, biofouling coverage can increase rapidly, and the onset can vary from year to year, even within farms from the same region. This makes it difficult for planning of harvest timing, and optimising biomass yield before quality is compromised (Njåstad, 2018).

The life cycle of *M. membranacea* and *E. pilosa* consists of two distinct life stages. One as an adult colony that forms encrusting layers on ephemeral substrates, such as kelp laminae, and another one as a microscopic planktotrophic larva, the cyphonautes, which develops through sexual reproduction (Woollacott and Zimmer, 2013). After sexual reproduction, larval populations of *M. membranacea* and *E. pilosa* cyphonautes feed on phytoplankton and are, therefore, able to remain in the water for days or weeks before settling on the kelp (Woollacott and Zimmer, 2013). After finding a suitable substrate, the larva settles and undergoes metamorphosis into a founding zooid (ancestrula), which gives rise to the colony through rapid asexual budding (Woollacott and Zimmer, 2013). Because cyphonautes of *M. membranacea* and *E. pilosa* are planktotrophic, they continue to grow while in the water column, and larval size increases as individuals progress toward metamorphic competence (Woollacott and Zimmer, 2013). Larger larvae are generally older, more developed, and more likely to be capable of settlement (Atkins, 1955; Ryland and Stebbing, 1971; Saunders and Metaxas, 2010). Larval size can, therefore, act as a proxy for both developmental stage and settlement potential, linking conditions in the plankton to subsequent biofouling pressure on kelp. A yearly monitoring program that quantifies cyphonautes abundance, and maturation state in the waters and in the kelp lamina, and also relates their presence with other environmental factors, such as temperature changes and the timing and composition of phytoplankton blooms, could provide an early warning for predicting when biofouling will start (Cohen, 2019; Førde et al., 2016; Matsson et al., 2019).

Temperature and water column stratification have been suggested as the main factors that increase growth rates of bryozoan colonies (Saunders and Metaxas, 2010). Light triggers photosynthesis and growth of phytoplankton, the main food source of bryozoan larvae, and has been suggested also to induce larval release from adult colonies (Woollacott and Zimmer, 2013). Other environmental factors and cues, such as food availability (using phytoplankton abundance as a proxy) (Woollacott and Zimmer, 2013), chemical signalling from bacterial biofilms and spectral changes in light might also play a role in larval settlement. Chemical cues released by macroalgae have been suggested to induce the transition from cyphonaute larvae of *M. membranacea* in the water column and settlement to the benthic habitat (Seed and O'Connor, 1981). The timing and concentration of the spring phytoplankton bloom likely influences cyphonautes abundance, settlement and colony growth. Likewise, phytoplankton species composition, specifically dominated by small flagellates, has been shown to increase bryozoan coverage on kelp and other substrates (Bullivant, 1968; Njåstad, 2018). Yet, it is unclear how these factors interact and trigger the sudden settlement and growth of bryozoans on lamina (days to a few weeks).

This paper aims to understand the factors that influence bryozoan colonies appearance and growth on a kelp farm from central Norway. The questions we aimed to answer with this study were: Does the timing

and size composition of the phytoplankton blooms influence the start of biofouling? Can larval abundance and size serve as a proxy for early biofouling detection? In this study, we hypothesized that the timing and size structure of the spring phytoplankton bloom, together with key environmental conditions, such as temperature and stratification, influence cyphonautes abundance and development. We further expected that larval abundance and size would predict the onset of bryozoan settlement on kelp. The results of this study highlight the environmental drivers of bryozoan appearance and growth on kelp, offering information that can enhance farm management and reduce biofouling-related losses.

2. Material and methods

2.1. Study site

The present study was conducted at a kelp farm at Frøya island, off the coast of Central Norway, Trøndelag (Fig. 1a), from February to June 2022 and 2023. The farm (Måsskjæra) belongs to Seaweed Solutions (SES) (63°44.617'N 8°52.756'E). The fieldwork was done at two sites, one by the edge of the seaweed farm (station MI) and another outside (station MO), approximately 500–700 m from MI (Fig. 1b and c). While the location of station MO was the same during both years, the location of station MI at the edge of the cultivation frame moved ~100 m from 2022 to 2023 due to farm logistics (Fig. 1c). The reason we chose these sites is that we wanted to collect larval samples from stations at the kelp cultivation site as well as a ~500–700 m away, to evaluate the temporal (stations MI and MO and 2 years) variability of the larvae in the kelp farm surroundings.

The Frohavet area is known to be biologically productive with intensive fishing and aquaculture activities (Ervik et al., 2018). The oceanography in the area is characterized by complex currents due to strong winds, shallow and uneven bathymetry, and water mixing events (from tides and storms) (Sætre, 2007). Two major currents are found along the coast of Norway - the North Atlantic Current (NAC) - a warm and nutrient-rich water mass that lies beneath the Norwegian Coastal Current (NCC) - a relatively low salinity water mass (Sætre, 2007). The NCC is comprised by freshwater runoff from fjords (including the Trondheimsfjord) that accumulates along the coast of Norway, particularly from spring to autumn. The seabed topography is comprised of mixed substrates, with large rock structures separated by areas with soft sediments (Keeley et al., 2019).

Måsskjæra is a semi-exposed kelp farm anchored with depth ranging from 8 to 35 m. The location is exposed to north-easterly winds and sheltered from westerly and southerly winds (Førde et al., 2016). The cultivation site is approximately 440 × 440 m and is based on a horizontal longline system divided into approximately 16 cultivation squares (Overrein et al., 2024) (Fig. 1c). *Saccharina latissima* (sugar kelp) is one of the main cultivated species at the SES farm and this species was seeded onto a 2 mm twine (polyester) at SES hatchery in Trondheim. Seedlings were deployed with a size of 1 cm. The seeded lines were spun around a 20 m stretch of carrying rope of 14 mm diameter. The distances between each line were 1.5 m. Måsskjæra is a commercial farm, but a designated grid at station MI was allocated specifically for research activities. At this station, seedlings from the two cultivation periods (October and January) were placed a few meters apart from each other.

2.2. Environmental data collection using sensors

Environmental data were collected to investigate if, and to what extent, they affect cyphonautes abundances. To record the physical properties of the water column at each station, vertical profiles (down to 10–15 m) of conductivity (later converted to salinity), temperature (°C), turbidity (Turb_{CTD}, Formazin Turbidity Units (FTU)) and chlorophyll *a* fluorescence (Chl_a_{CTD}, mg m⁻³) were performed at each station (MI and MO) using a CTD device (model SD204 SAIV A/S™). Dates for CTD

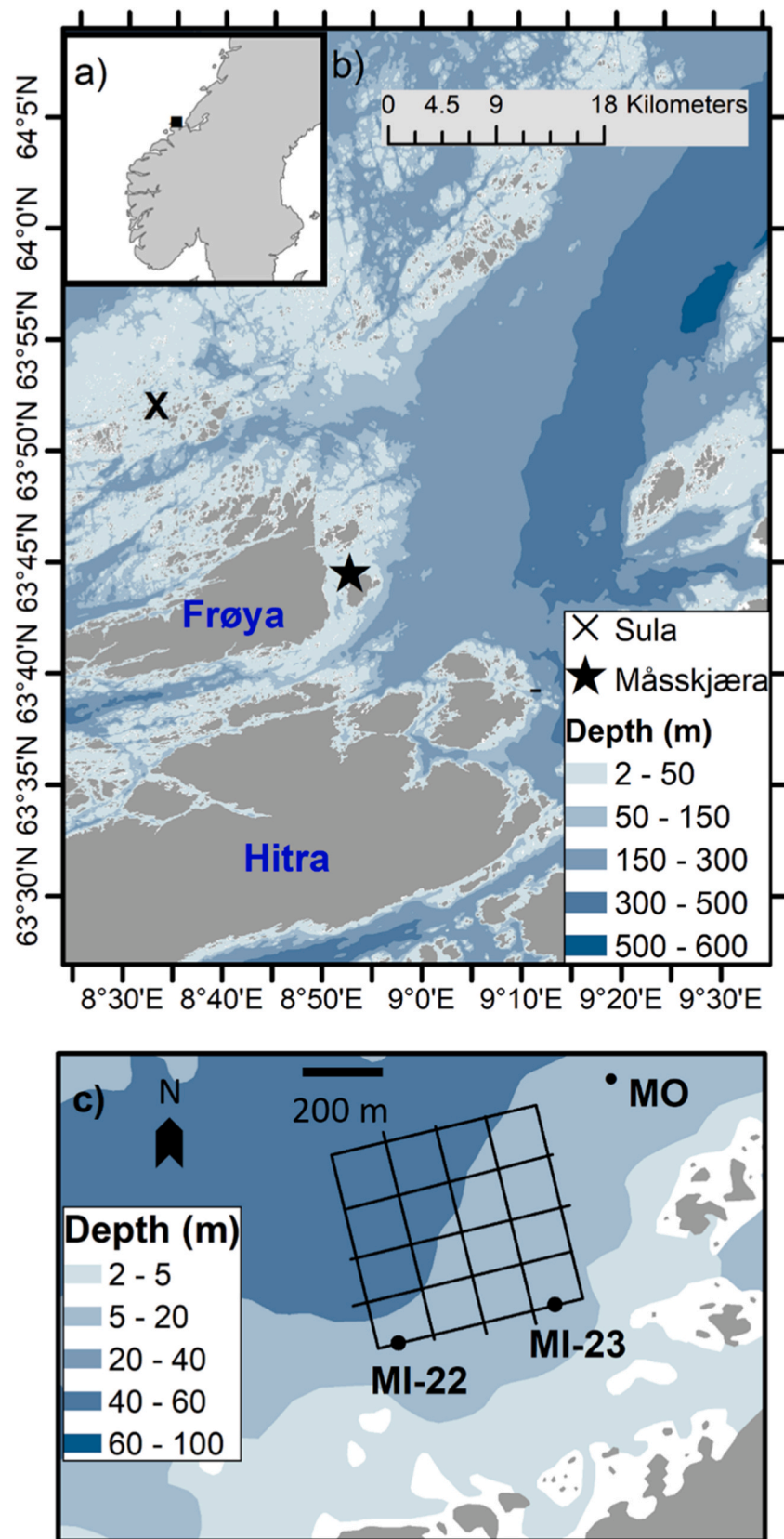


Fig. 1. a) Map of the geographical location of Måsskjæra kelp farm located off the coast of central Norway. b) The location of the seaweed farm (star symbol), including Sula meteorological station (X symbol), where wind data were collected. c) Illustrations of the layout of the seaweed farm and the sampling sites for environmental data and cyphonautes larvae abundance (Stations MI in 2022 and 2023 and MO – same station for 2022/3). The cultivation site in c) is approximately 440 × 440 m.

deployment as well as net and water samples are shown in Table 1.

In addition to vertical profiles with a CTD, a C3 submersible fluorometer sensor (Turner Designs, USA) was deployed at 3 m depth and attached to the mooring frame of the farm at station MI-22 in 2022 and MI-23 in 2023 (Fig. 1c). The sensor collected time-series data (every 10 min) of chlorophyll *a* fluorescence (Chl_a_{frame}, later converted to biomass values, mg m⁻³), turbidity (Turb_{frame}, FTU) and temperature (°C) from February to June in 2022 and 2023. The sensor had an antifouling copper plate and a mechanical wiper that rotated and cleaned the optical sensors before each measurement was taken.

To visualize the overall trend of above-water, solar radiation available throughout the cultivation season, satellite-derived all-sky surface total Photosynthetically Active Radiation (PAR_{SAT}, 400–700 nm) was obtained from the NASA Power's online database from Måsskjæra's site (NASA POWER, 2024; <https://power.larc.nasa.gov>). Wind speed data (daily average, m s⁻¹) was extracted from the Sula meteorological station in western Frohavet (Fig. 1b) and from the Norwegian Centre for Climate Services (<https://seklima.met.no/>).

2.3. Water and net sampling

Water and net sampling (more details below) were taken at both sites over the two years (Table 1). Sampling was done once or twice a month from February to June 2022 and 2023, adding up to 17 days in total (8 days in 2022 and 9 days in 2023). Selection of sampling days was

Table 1

Sampling dates for both years where CTD measurements (10 m) and water sampling for dissolved inorganic nutrients were performed (X). Sampling dates for collection of *S. latissima* specimens (3–5 replicates) for colony scanning for both the deployment lines (October 2022 and January 2023) are given.

Date	CTD and water and net sampling	Replicate nr. from the line deployed in Oct '22	Replicate nr. from the line deployed in Jan '23
February 16, 2022	X	–	–
March 22, 2022	X	–	–
April 05, 2022	X	–	–
April 20, 2022	X	–	–
May 04, 2022	X	–	–
May 20, 2022	X	–	–
June 03, 2022	X	–	–
June 15, 2022	X	–	–
February 15, 2023	X	–	–
March 16, 2023	X	–	–
March 29, 2023	X	–	–
April 18, 2023	X	–	–
April 25, 2023	X	3	–
May 10, 2023	X	5	5
May 15, 2023	–	5	5
May 23, 2023	X	5	5
June 07, 2023	X	5	5
June 15, 2023	–	5	5
June 29, 2023	X	5	5

partially dependent on weather and tidal conditions (preferably low tides and weak wind conditions for consistency), as well as personnel availability.

Seawater samples were collected with a custom-made 5 L water sampler, at 3 m depth at each site. The water was stored in two 8 L acid-washed dark bottles and filtered at the dock a few hours later for nutrients, and phytoplankton size structure.

For nutrient concentrations (nitrite plus nitrate [NO₂⁻ + NO₃⁻] - hereafter referred as to nitrate [NO₃⁻]) as well as ammonium [NH₄⁺], phosphate [PO₄³⁻] and silicate [Si(OH)₄], triplicate water samples from each station were filtered with a 0.8 µm polycarbonate filter (Whatman, International Ltd.), after being flushed to remove artificial NH₄⁺ residuals. The filtrate (~45 mL of each sample) was placed into centrifuge tubes and kept on dry ice for transportation and stored at –20 °C in the freezer until further analysis.

For phytoplankton size structure, triplicate water samples (~10 mL) were placed in centrifuge tubes and fixed with pre-filtered 25 % glutaraldehyde (Fisher Scientific U.K.) at a final concentration of 0.25 %. The centrifuge tubes were then wrapped in aluminum foil and after 15–30 min of fixation stored on dry ice until transportation to a –80 °C freezer at Trondheim Biological Station (TBS), according to Fragoso et al. (2019).

Zooplankton net sampling was conducted for estimating cyphonautes abundance. Vertical hauls of zooplankton from the surface (upper 5 m) were taken using a handheld zooplankton net (100 µm mesh size and 40 cm in diameter, Hydrobios) and a filtering cod-end. Duplicate samples were collected at 2022 and triplicate samples were collected at 2023. The net was rinsed with 96 % ethanol (EtOH) each time and were preserved in this solvent in 200 mL plastic bottles and stored dark and in room temperature for later analysis.

2.4. Kelp sampling for bryozoan colony identification

The quantification of bryozoan coverage on the kelp lamina from digital scans was done only from *S. latissima* specimens cultivated in 2023 from station MI-23. The reason for this is that a protocol for bryozoan detection was applied in 2022, but it was unsuccessful for identification of small colonies (Thomason, 2023), and, therefore discarded in this study. In 2023 a new methodology was developed, where cyphonautes larvae, ancestrula and small to large colonies in the kelp laminae were detected using a flatbed scanner (Snijder, 2024). For this, *S. latissima* specimens were collected from two distinct cultivation lines (seedlings deployed in October 2022 and January 2023) every 1–2 weeks at seven sampling dates subject to weather and boat availability from late April until the end of June 2023. Specifically, replicate samples of *S. latissima* sporophytes were collected at station MI across seven sampling days (25th April, 10th May, 15th May, 23rd May, 7th Jun, 15th Jun, 29th Jun) to register bryozoan coverage on the laminae. On every sampling date, five specimens from each deployment line were randomly selected (Table 1). The kelp samples were rolled up in aluminum foil, put into plastic bags, stored in dry ice (–80 °C) for transportation, and later stored in a freezer (–20 °C) at TBS, Trondheim, Norway.

2.5. Laboratory analysis of nutrient concentrations

Filtered seawater samples for [NO₃⁻], [Si(OH)₄], [PO₄³⁻] and [NH₄⁺] were analyzed at TBS. Analysis for dissolved nutrients in water samples were done by using Flow Solution IV system O.I. Analytical Auto analyzer according to the Norwegian standard 4724 and 4745 respectively. Determination of NH₄ was done by direct segment flow analysis according to (Kérouel and Aminot, 1997).

2.6. Phytoplankton size structure

Samples were analyzed within 24 months of collection using a

CytoSense benchtop flow cytometer (CytoBuoy, b. v., NL) equipped with a blue laser (488 nm, 50 mW) to allow discrimination between phototrophic and non-phototrophic particles. Specifications about the sheath fluid are found in Fragoso et al. (2019). Yellow-green (505/515) fluorescence beads (Molecular Probes, 1 μm) were used for size calibration before few consecutive runs. Fixed samples were transferred to a glass beaker and kept in suspension using a magnetic stirrer and injected into the CytoSense at a particle rate below 5000 particles s^{-1} . Each sample was run under two different protocols based on phytoplankton size: one to detect pico- and nanophytoplankton communities and another one for microphytoplankton, according to Hubert et al. (2024).

Manual clustering (or ‘gating’ in flow cytometry studies) and identification of the main phytoplankton groups occurred based on light scattering and fluorescence properties of the particles, and in accordance to the interoperable vocabulary (Thyssen et al., 2022). Cytogram analyses were performed using CytoClus 4 software (CytoBuoy b. v., Netherlands). For simplicity, grouping of the phytoplankton functional groups were divided into size categories: pico- ($\sim < 2 \mu\text{m}$), nano- ($\sim 2\text{--}20 \mu\text{m}$), and microphytoplankton ($> 20 \mu\text{m}$).

2.7. Cyphonautes abundance from net samples

Identification and quantification of *M. membranacea* and *E. pilosa* cyphonautes from plankton net samples were performed at TBS using a Leica (model M205C) dissecting microscope. The EtOH-preserved samples were drained over a 110 μm mesh and rinsed with filtered seawater. After rinsing, the samples were flushed into a plastic beaker for counting and diluted to a known volume. Subsamples of 3.5 mL were pipetted into a plexiglass counting chamber with four grooves, and the number of *M. membranacea* and *E. pilosa* larvae were counted. Subsamples were taken from each sample until 100 individuals of at least one of the species had been counted, or until the whole sample had been counted.

2.8. Cyphonautes size measurements from images

For cyphonautes size measurements, fixed zooplankton net samples were run using an imaging-recording microscope, called the AFTI-scope (Fig. 2a). The AFTI-scope is a flow through, open-source microscope designed to automatically capture images of particles in water samples inspired by the PlanktoScope design (Pollina et al., 2022). For more details about the AFTI-scope and how it was used to analyze cyphonautes larvae (Hama, 2024; Haugum et al., 2023). A total of 21 net samples - triplicates of 7 sampling dates at station MI from February to early June 2023 were run through the AFTI-scope after microscopic counts. Although samples from late June 2023 were counted in the stereomicroscope, it was not possible to run them through the AFTI and collect their size measurements. This was due to high densities of the dinoflagellate *Tripos* spp. and clogging in the flow-through AFTI system. Yet, analyzing samples from February to early June gave the opportunity to compare the cyphonautes size during the kelp cultivation period.

Before running the preserved net samples in the AFTI-scope, samples were rinsed using a 110 μm mesh to remove the EtOH and placed into a plastic beaker with filtered seawater with a known volume. All sample volumes in filter were run through the AFTI-scope. After the imaging record, images were analyzed and manual annotation of length and width of larvae of *M. membranacea* or *E. pilosa* were performed using the image processing program ImageJ (version 1.53 k). Using the ‘line’ tool in ImageJ, two lines were manually drawn in each larvae picture: one representing the antero-posterior basal axis (basal edge) and another one from oral to aboral axis (length) (Fig. 2b). An image of a microscopic ruler taken with the AFTI-scope in the same setup was used as size reference to set a scale of pixels per centimeter (pixels cm^{-1}) for the corresponding frame. A total of 390 cyphonautes were measured.

2.9. Flatbed image scanning of kelp lamina

To get highly resolved images of the kelp lamina (Fig. 2c–h), a Canon flatbed scanner LiDE 400 with a scan resolution of 1200 DPI (=47.24 Dots per mm) was used (Fig. 2d). An adhesive measurement tape was attached to the inside part of the scanner to generate a scale bar (Fig. 2e). Scans were made with the Canon IJ Scan Utility software, and the DPI was set to 1200 for high image resolution to detect the small bryozoan colonies, settlers and ancestrulae (minimum size diameter of 300 μm on the kelp lamina) (Fig. 2f–h). The files were stored as TIFF and later processed into SVG files. For image acquisition, frozen specimens of *S. latissima* were thawed before scanning and cut into small parts ($< 20 \text{ cm}$) so that the lamina pieces could fit into the scanner (Fig. 2c). Despite its advantages, such as low costs and high resolution, the flatbed method may introduce minor blade deformation and can be time-consuming, limiting the amount of kelp lamina to be analyzed. More details about the scanning process are described in Snijder (2024).

2.10. Bryozoan image processing

A total of 1037 image scans, including both sides of the lamina, were made from 63 specimens. Bryozoan colonies, larval settlement and pre-ancestrulae on the images were manually marked with circles in the vector graphics editor program Inkscape. The pixel size of the kelp lamina and the marked bryozoan area (larvae, pre-ancestrula or colony) from the scans were calculated in the programming language Python™ (latest version used: 3.12.3 (Van Rossum and Drake Jr, 1995)). The lamina area from the scans were retrieved by using the OpenCV library in Python (Bradski, 2000), where every scanned image was converted into grayscale and a threshold was made to create a binary mask, so the pixels of the lamina could be counted. Counts of the bryozoan colonies, settled larvae and pre-ancestrulae were also retrieved using Python, where every circle drawn in Inkscape was counted as one. For normalizing the data, the total counts were divided into the total scanned lamina surface area and calculated to counts per square meter area. For more details, see Snijder (2024).

2.11. Statistical analysis

All statistical analyses and graphs were conducted in Minitab 22.2.2 and Matlab R2024a using the *Statistics* and *Machine Learning Toolbox*. Cyphonautes larvae and phytoplankton abundances were checked for normality using a Shapiro-Wilk test and differences ($p < 0.05$) among years and stations were either tested using the two-sample *t*-test or Mann-Whitney, depending on normality. Boxplots indicate the median, the 25th and 75th percentiles, and outliers.

To assess the environmental drivers of larval abundance, size, and colony counts of *E. pilosa* and *M. membranacea* in the water column and on kelp, we used generalized linear mixed-effects models (GLMMs) with a significance threshold of $p = 0.05$. GLMMs were fitted separately for larval abundance, larval size, and colony counts based on what type of data that was collected simultaneously and can be paired. For instance, larval abundance (from net samples) was modeled using environmental predictors measured concurrently from water samples, CTD data (value from 3 m depth – same depth as kelp cultivation) and time series data. From water samples the predictors were PO_4^{3-} , NH_4^+ , NO_3^- , Si(OH)_4 , and phytoplankton size classes - pico-, nano-, microphytoplankton. From CTD data predictors were temperature (T_{CTD}), salinity (S_{CTD}), $\text{Chl}a_{\text{CTD}}$ and Turb_{CTD} . For larval abundance models, we retained all environmental predictors even when collinearity was present, since each variable represents a distinct ecological process potentially influencing larval supply. This exploratory approach allowed us to test a broad suite of environmental associations, while acknowledging that collinearity may reduce the precision of individual coefficient estimates. For larval size models, we reduced the number of predictors to 7 compared to the abundance models. This was done to avoid overfitting, given the smaller

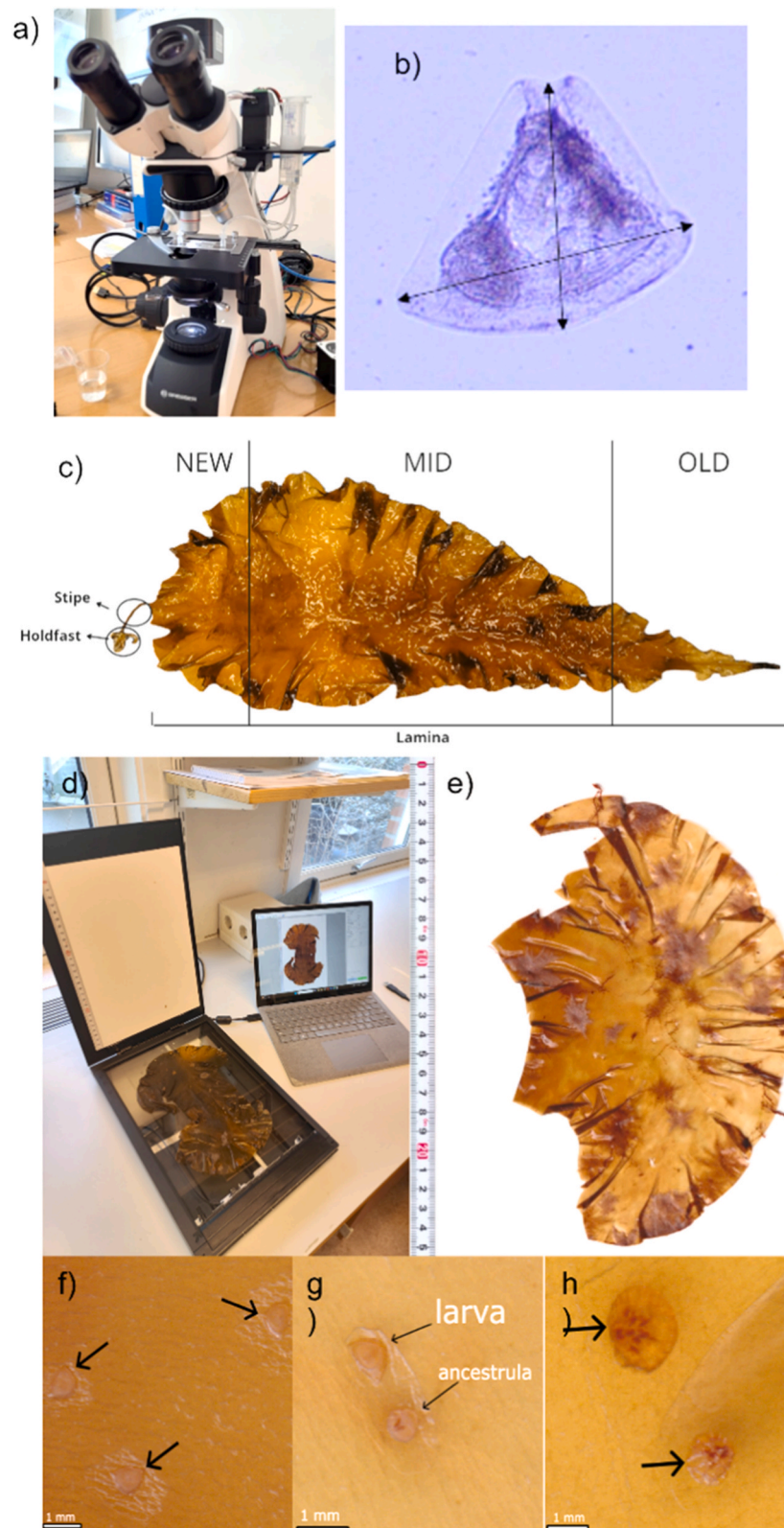


Fig. 2. a) The AFTI-scope scope assembled on the Bresser Trino TFM-301 microscope. b) Lines showing the size measurements (basal edge and length) extracted from cyphonautes larvae individuals – antero-posterior basal axis (basal edge, horizontal line) and oral-aboral axis (length, vertical line). c) An image of a sporophyte with lamina, stipe and holdfast and an example showing the divisions of the kelp lamina into new (meristem), middle and old tissue (tip). d) The Canon A4 paper-size flatbed scanner LiDE 400 with kelp and (e) one example of an image with a scale bar after scanning. f) Examples of three settled larvae (note the triangular shell shape), g) a settled larva and a pre-ancestrula stage, where the septum is clearly observed and h) a developed small colony (approximate spatial resolution of 21 μm).

sample size for size measurements (9 samples-only stations MI from 2023 had size measurements), and to focus on environmental variables most likely to affect larval growth (T_{CTD} , S_{CTD} , $Chla_{CTD}$, $Turb_{CTD}$, wind, and PAR_{SAT}). Nutrients and phytoplankton groups were excluded here because their effects are expected to act indirectly larval size through food availability, which was already represented by chlorophyll *a*.

Colony counts on kelp panels were analyzed with GLMMs assuming a Poisson distribution and log link, including environmental predictors and deployment as fixed effects, and specimen ID as a random intercept. Predictors were deployment (categorical; month of deployment), T_{frame} , $Turb_{frame}$, and $Chla_{frame}$. We avoided including concurrent larval metrics to prevent post-treatment bias, and we limited the predictor set to reduce collinearity and overfitting. All models included random intercepts for Station, Year, or deployment as appropriate. Seasonal variation was modeled using sine and cosine transformations of day-of-year (YD). Model predictions were compared to observed values using scatterplots and seasonal trends. Predicted and observed larval abundances were smoothed with LOWESS and visualized separately for 2022 and 2023.

3. Results

3.1. Time-series of environmental data

In both years, the average daily wind speed was high (many days above 10 m s^{-1}) in winter months, reducing to below 10 m s^{-1} from April/May, with except of few storm events in spring (Fig. 3a). This coincided with the continuous increase in photoperiod (increased daylight period from 100 to $200 \mu\text{mol photons m}^{-2} \text{ s}^{-1}$ average daily light intensity (in PAR_{SAT} in March), sea surface temperature (SST $>6^\circ\text{C}$), the start of the spring phytoplankton bloom (values $>2 \text{ mg m}^{-3}$) (Fig. 3b–d) and water column stratification (Fig. S1a–d, supplementary material). Salinity varied between sites and with time, being generally fresher at station MI than at MO and in 2022 than in 2023, particularly in late March and mid-May (salinity down to 32) (Fig. S1e–h, supplementary material). Storm events in spring, inferred by an increase in average daily wind speed again ($>10 \text{ m s}^{-1}$) and average reduction in average daylight due to clouds, occurred in the beginning of May in 2022 and in the end of May/beginning of June 2023 (Fig. 3a and b).

Calm winds, in addition to increased daily PAR_{SAT} and increased sea surface temperatures (SST $>5^\circ\text{C}$) in April 2022 and 2023, appeared to be major meteorological factors that initiated the spring phytoplankton

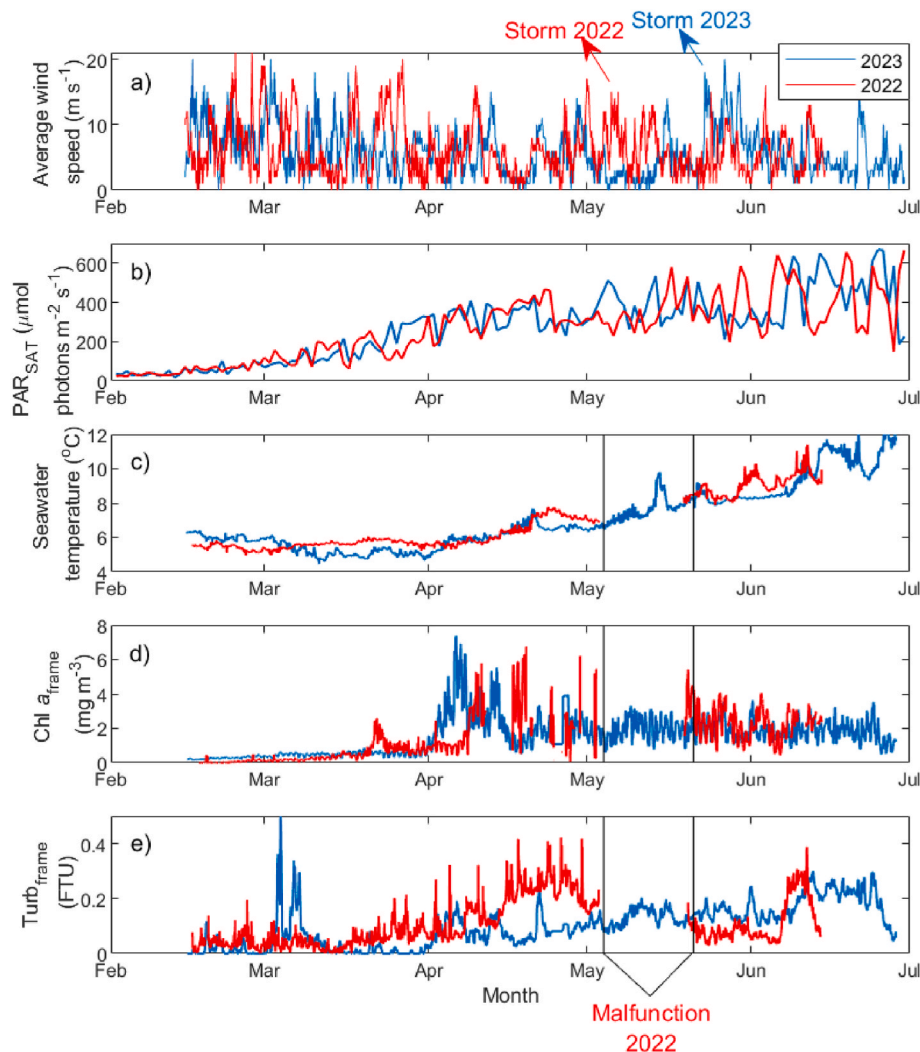


Fig. 3. a) Time series data of a) average daily wind speed (m s^{-1}), b) average daily irradiance (PAR_{SAT} , $\mu\text{mol photons m}^{-2} \text{ s}^{-1}$), c) seawater temperature ($^\circ\text{C}$), d) chlorophyll *a* fluorescence ($Chla_{frame}$, mg m^{-3}) and e) turbidity ($Turb_{frame}$, FTU) collected every 10-min using the C3 submersible fluorometer at Måsskjæra farm (station MI, inside the farm) from mid-February to mid- (2022, red) and end of June (2023, blue). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

bloom (Fig. 3a–d). Although the main phytoplankton peak ($\sim 5\text{--}6\text{ mg m}^{-3}$ $\text{Chla}_{\text{frame}}$ concentration) was in April, a single peak was observed in 2023, while two bloom peaks were observed in 2022 - a smaller one in the end of March ($\sim 2\text{ mg m}^{-3}$), followed by a larger one (~ 2 weeks) in mid-April (up to 6 mg m^{-3}) (Fig. 3d). From mid-May to mid-June, the $\text{Chla}_{\text{frame}}$ concentration in the water appeared to be $\sim 2\text{ mg m}^{-3}$ in both years, reflecting the high productivity of the area (Fig. 3d).

After the spring bloom peak in April, SST increases gradually until the end of June (from 6°C to approximately $10\text{--}12^\circ\text{C}$), showing some peaks in warming periods, most related to the influence of weaker winds (Fig. 3c). Surface water was generally colder in 2023 when compared to 2022 by $1\text{--}2^\circ\text{C}$.

Values for $\text{Turb}_{\text{frame}}$ varied between years, being higher in 2022 and reflecting the peak of the spring bloom in 2022 (0.35 FTU), while it peaked in March 2023 ($>0.4\text{ FTU}$) and increased from April to July (from 0.1 to 0.2 FTU), reflecting not only phytoplankton but other particles in the waters (marine snow and sediments) (Fig. 3e).

3.2. Nutrient concentrations

Nutrient concentrations did not significantly vary between years and sites ($p > 0.05$, t -test), except for NH_4^+ ($p = 0.0001$, Mann-Whitney test), which was higher in 2022 (average = $0.8\text{ }\mu\text{M}$) than in 2023 (average =

$0.2\text{ }\mu\text{M}$). Nutrient concentrations followed a general trend typically observed for temperate Northeastern Atlantic waters (except for NH_4^+ in 2023), being high during February, and decreasing with time (particularly during the spring bloom in the end of April) (Fig. 4). After the first bloom peak in April, nutrients bounced slightly back up in May 2022 and June 2023 (Fig. 4), possibly because of strong winds and mixing (see Fig. 3a). In general, NO_3^- concentrations were slightly higher in station MO (up to $4\text{ }\mu\text{M}$ higher in some cases) compared to MI and in 2022 (up to $10\text{ }\mu\text{M}$) and compared to 2023 (up to $5\text{ }\mu\text{M}$) (Fig. 4a and b). For PO_4^{3-} concentrations, a similar trend to NO_3^- was observed, being slightly higher at stations MO (up to $0.1\text{ }\mu\text{M}$ higher) than station MI (Fig. 4c and d). During winter 2022 (February and March), NH_4^+ concentrations were exceptionally high ($>1\text{ }\mu\text{M}$) (Fig. 4e and f). SiOH_4 concentrations followed similar patterns when compared to PO_4^{3-} and NO_3^- , being high before the bloom ($\sim 3\text{ }\mu\text{M}$), at a storm event in May 2022 and June 2023 ($\sim 2\text{ }\mu\text{M}$) and station MO (up to $0.1\text{ }\mu\text{M}$ higher) than station MI (Fig. 4g and h).

3.3. Phytoplankton and cyphonautes abundance and size distributions and recruitment

Related to phytoplankton size-structure, abundances of pico-, nano- and microphytoplankton were not significantly different between years

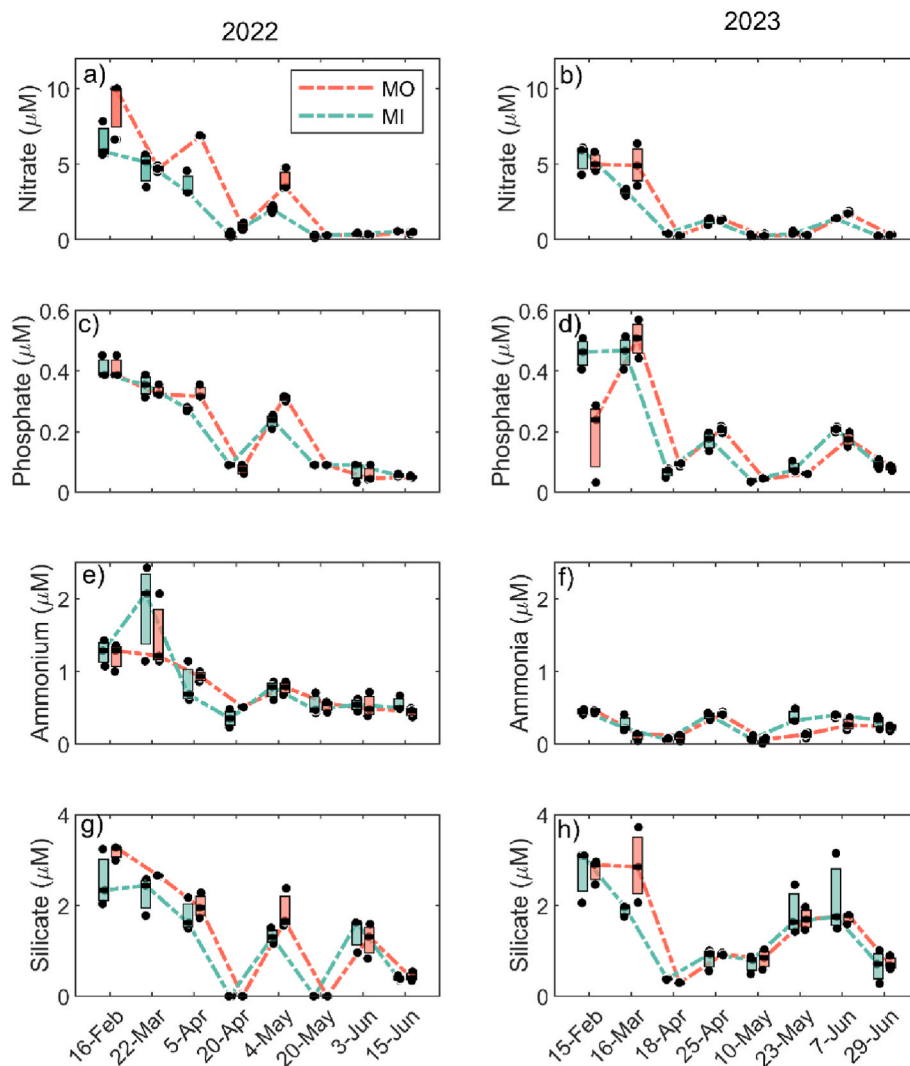


Fig. 4. Nutrient concentrations (nitrate, phosphate, ammonium and silicate in μM) collected every 2 weeks from February to June in 2022 (left) and 2023 (right) from MI (green) and MO (orange) stations. Dots represent replicates. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

and sites ($p > 0.05$, two-samples T-test). In general phytoplankton abundances were as expected low in winter months ($<40 \times 10^6$ ind m^{-3} in February and early March), increasing with time and peaking on June 3, 2022 at stations MI and MO and May 23, 2023 (abundances $>100 \times 10^6$ ind m^{-3}) (Fig. 5a and b). Picophytoplankton and nanoplankton were more abundant from late May until late June in 2022 and 2023. Picoplankton contributed to the largest fraction of the phytoplankton, reaching $>65\%$ on June 3, 2022 (for both MI and MO stations), followed by nanophytoplankton $>58\%$ on April 20, 2022 and April 18, 2023 (particularly at station MI) (Fig. 5a and b). The microphytoplankton population contributed to an average of 13 % of phytoplankton for both locations and years, peaking on March 22, 2022 and April 18, 2023 (Fig. 5a and b).

Abundance of cyphonautes larvae from stations MI and MO was highly variable among replicates (Fig. 5c and d), although not significantly different among years for *M. membranacea* and *E. pilosa* ($p = 0.635$ and $p = 0.184$, respectively, two-samples T-test) and stations ($p = 0.725$ and $p = 0.932$, respectively, two-samples T-test). However, a general trend was observed with *M. membranacea* larval stages in higher concentrations in most cases, particularly in 2022 (Fig. 5e). Two peaks in abundance of cyphonautes were observed for *M. membranacea* in both years: early April (up to 177 ind m^{-3} in station MI) and early June in 2022 (values up to 341 ind m^{-3} at station MO), while in 2023, the peak was observed in mid-to late April (values for *E. pilosa* cyphonautes up to 403 ind m^{-3}) and late June in 2023 (values up to 444 ind m^{-3} for *M. membranacea*) (Fig. 5c and d).

Cyphonautes were also highly variable in size, ranging from 0.07 to 0.417 mm in length, respectively for *E. pilosa* larval stages ($n = 165$) (Fig. 5e). For *M. membranacea* larval stages, the sizes ranged from 0.075 to 0.605 mm for length ($n = 225$) (Fig. 5e). There was a general increase in size from late May until late June, particularly for *M. membranacea* cyphonautes (Fig. 5e). The median value of length was ~ 0.15 mm in

mid-March ($n = 22$), while in early June, these values were 0.52 mm ($n = 30$), respectively (Fig. 5e). In late June, however, the values of length ($n = 10$) were reduced to 0.35 mm, respectively (Fig. 5e). Bryozoan total colony area increased in time, particularly after May (Fig. 5f). The largest colonies (between 10^3 to 10^4 mm 2) were only found on the June 29, 2023 (Fig. 5f).

Saccharina latissima from both deployment lines had small percentage bryozoan coverage (10^{-5} to 10^{-2} % cover) from late April to early May (Fig. S2a, supplementary material). However, the percentage coverage increased rapidly in the middle of June, with a higher percentage of coverage for the October deployment line (average of 11 % coverage) than the January deployment (average of 3 % coverage) on June 29, 2023 (Fig. S2a, supplementary material). In total, 1.3307×10^5 colonies of settled larvae and pre-ancestrulae were detected on the scanned images, from which 3803 counts were from the replicates from the line deployed in January 2023 and 9.504×10^4 counts from the line deployed in October 2022. The first settlement was detected in the first kelp sampling on April 25, 2023 from the October deployment, with a total of 6 settled larvae from samples taken that day. No clear colonies were detected at that stage (Fig. S2b and c, supplementary material). Two peaks in recruitment occurred for both lines on the May 15, 2023 and the June 15, 2023 (187 and 826 counts per kelp m^{-2} for October, respectively, and 126 and 618 counts per kelp m^{-2} for January) (Fig. S2b, supplementary material). Both coverage and colony abundances also increased exponentially with time. Almost all settlers and pre-ancestrulae were found on the meristem, the newer part of the lamina, for the whole sampling period (Snijder, personal observation). The total counts of bryozoan colonies m^{-2} in *S. latissima* increased gradually and showed the same pattern of variation for both deployment lines, with the highest number of colonies on June 29, 2023 (approximately 1260 counts per m^2 for October and 860 for January, Fig. S2c, supplementary material). For June, bigger colonies (e.g. > 5 cm) were

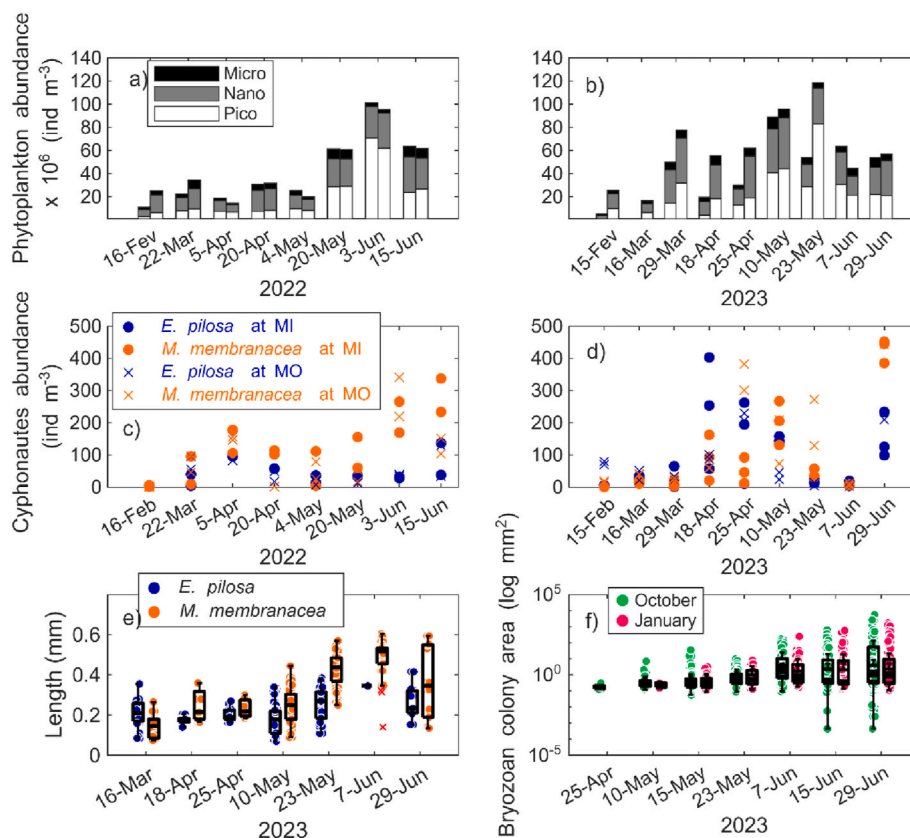


Fig. 5. Phytoplankton ($\times 10^6$ ind m^{-3}) and cyphonautes larval abundance (ind m^{-3}) in a,c) 2022 and b,d) 2023; e) larval length (mm); and f) bryozoan total colony area in 2023 (log mm 2 , station MI only). For each sampling day in panels a,b, the first bar corresponds to MI and the second to MO.

mainly found at the older part (tip) of the lamina. The highest abundance of colony sizes was found in the smallest size range (0–0.5 log mm²) for this date as well (Fig. S2d, supplementary material).

3.4. General linear mixed models (GLMM)

For *E. pilosa*, larval abundance showed clear seasonal variation (SinYD, $\beta = 5.10 \pm 1.32$, CosYD, $\beta = -2.34 \pm 0.67$). Negative associations were also detected between *E. pilosa* larval abundance and PO₄³⁻ ($\beta = -6.95 \pm 1.78$, $p = 0.0002$), S_{CTD} ($\beta = -2.06 \pm 0.52$, $p = 0.0002$) and PAR_{SAT} ($\beta = -0.007 \pm 0.002$, $p = 0.003$), while a weaker negative effect was seen for NH₄⁺ ($\beta = -1.32 \pm 0.41$, $p = 0.002$) and pico-phytoplankton ($\beta = -0.03 \pm 0.01$, $p = 0.01$) (Table 2).

For *M. membranacea*, similar seasonal patterns were evident (SinYD: $\beta = 6.67 \pm 1.44$, 0.00002; CosYD: $\beta = -2.04 \pm 0.73$, $p = 0.007$) (Table 2). Among environmental predictors, S_{CTD} ($\beta = -2.29 \pm 0.56$, $p = 0.0001$), NH₄⁺ ($\beta = -0.92 \pm 0.45$, $p = 0.04$), and PAR_{SAT} ($\beta = -0.005 \pm 0.002$, $p = 0.04$) also had a significant negative association, while T_{CTD} (0.67 ± 0.29 , $p = 0.02$) had a positive association with larval abundance of *M. membranacea*. Correlations between phytoplankton size structure (pico-, nano-, and microphytoplankton) and cyphonautes abundance for both species were weak ($r < 0.13$, data not shown).

Larval size models revealed no significant predictors for *E. pilosa* (Table 2). In contrast, *M. membranacea* larval size decreased with salinity ($\beta = -1.58 \pm 0.63$, $p = 0.01$), Chl_a_{frame} ($\beta = -0.97 \pm 0.27$, $p = 0.0004$), and wind ($\beta = -0.13 \pm 0.05$, $p = 0.005$), while it increased with Turb_{frame} ($\beta = 7.34 \pm 1.67$, $p = 0.00002$) and light (PAR_{SAT}: $\beta = 0.006 \pm 0.002$, $p = 0.00008$) (Table 2).

For bryozoan settlement and colony counts, deployment timing ($\beta =$

0.20 ± 0.01 , $p < 0.0001$), temperature ($\beta = 0.93 \pm 0.10$, $p < 0.0001$), and chlorophyll *a* ($\beta = 1.14 \pm 0.47$, $p = 0.01$) were the strongest correlations (Table 2). Additional information about GLMM are found in Table S1 (supplementary material).

According to the predicted and observed trends of larval abundances, both bryozoan species showed two seasonal pulses in larval abundance. In 2022, *E. pilosa* peaked in early April (max ~ 200 ind m⁻³ in Apr; ln (count+1) = 5.18) and again in mid-June (max ~ 400 ind m⁻³ in mid-June; ln = 4.93), while *M. membranacea* peaked strongly in June (~600 ind m⁻³ on Jun 3; ln = 5.84) after a smaller April pulse (~400 ind m⁻³, Fig. 6a and b). In 2023, *E. pilosa* reached a pronounced April peak (max ~ 400 ind m⁻³ on 18-Apr; ln = 6.00) with a secondary pulse in late June (up to 233 ind m⁻³ on Jun 29; ln = 5.46). *M. membranacea* peaked in late April (up to ~400 ind m⁻³ in late Apr; ln = 5.95) and again in late June (max ~ 600 ind m⁻³ on 29-Jun; ln = 6.45). Related to larval size, highest values peaks in abundance were found in late May and early June compared to February (from 0.2 to 0.5 mm) (Fig. 6c). The kelp deployed in October had higher concentration of bryozoan colonies than kelp deployed in January (Fig. 6d).

4. Discussion

4.1. Phytoplankton and cyphonaute larvae dynamics

The onset of spring phytoplankton blooms in the Frohavet region are affected by weather, particularly calm winds and cloud-free days, which reduces the mixed layer depth and increase light availability for phytoplankton growth (Fragoso et al., 2024). In this study, the phytoplankton bloom started earlier in 2022 (late March) than in 2023 (early

Table 2
Results from the general linear mixed models. ** $p < 0.001$; * $p < 0.05$

<i>E. pilosa</i> larvae abundance. AIC: 222.23, BIC: 266.77, LogLikelihood: 92.117, Deviance: 184.23.					<i>M. membranacea</i> larvae abundance. AIC: 236.01, BIC: 280.54. LogLikelihood: 99.006, Deviance: 198.01.				
Term	Est.	SE	DF	pValue	Est.	SE	DF	pValue	
Intercept	72.33	19.00	61	**0.0003	72.59	20.77	61	**0.0009	
SinYD	5.10	1.32	61	**0.0003	6.67	1.44	61	**0.00002	
CosYD	-2.34	0.67	61	**0.0008	-2.04	0.73	61	**0.007	
Pico	-0.03	0.01	61	*0.01	0.01	0.01	61	0.3	
Nano	0.03	0.02	61	0.2	0.04	0.03	61	0.1	
Micro	0.01	0.08	61	0.9	-0.02	0.09	61	0.8	
PO ₄ ³⁻	-6.95	1.78	61	**0.0002	-3.19	1.94	61	0.1	
NH ₄ ⁺	-1.32	0.41	61	*0.002	-0.92	0.45	61	*0.04	
NO ₃ ⁻	0.18	0.15	61	0.2	0.28	0.16	61	0.1	
SiOH ₄	-0.17	0.29	61	0.5	-0.62	0.31	61	0.1	
T _{CTD}	0.10	0.26	61	0.7	0.67	0.29	61	*0.02	
S _{CTD}	-2.06	0.52	61	**0.0002	-2.29	0.56	61	**0.0001	
Chl _a _{CTD}	-0.10	0.13	61	0.5	-0.20	0.14	61	0.2	
Turb _{CTD}	-5.12	2.54	61	*0.05	-1.51	2.78	61	0.6	
Wind	0.02	0.09	61	0.8	0.08	0.10	61	0.4	
PAR _{SAT}	-0.01	0.002	61	*0.003	-0.01	0.002	61	*0.04	
<i>E. pilosa</i> larvae size AIC: 222.23, BIC: 266.77 LogLikelihood: 92.117, Deviance: 184.23.					<i>M. membranacea</i> larvae size AIC: 457.55, BIC: 426.8 LogLikelihood: 237.77, Deviance: 475.55				
Term	Estimate	SE	DF	pValue	Estimate	SE	DF	pValue	
Intercept	-13.93	26.33	158	0.6	52.00	20.92	218	*0.01	
T _{CTD}	-0.06	0.11	158	0.6	0.23	0.09	218	*0.007	
S _{CTD}	0.41	0.80	158	0.6	-1.58	0.63	218	*0.01	
Chl _a _{CTD}	-0.04	0.32	158	0.9	-0.97	0.27	218	**0.0004	
Turb _{CTD}	2.44	1.56	158	0.1	7.34	1.67	218	**0.00002	
Wind	0.05	0.05	158	0.3	-0.13	0.05	218	*0.005	
PAR _{SAT}	0.001	0.002	158	0.7	0.01	0.002	218	**0.00008	
Colony/recruitment counts kelp m ⁻² AIC: 457.55, BIC: 426.8, LogLikelihood: 237.77, Deviance: 475.55									
Term	Estimate	SE	DF	pValue					
Intercept	-5.05	1.27	58	**0.0002					
Deployment	0.20	0.01	58	**<0.0001					
T _{frame}	0.93	0.10	58	**<0.0001					
Turb _{frame}	-0.40	4.41	58	0.9					
Chl _a _{frame}	1.14	0.47	58	*0.01					

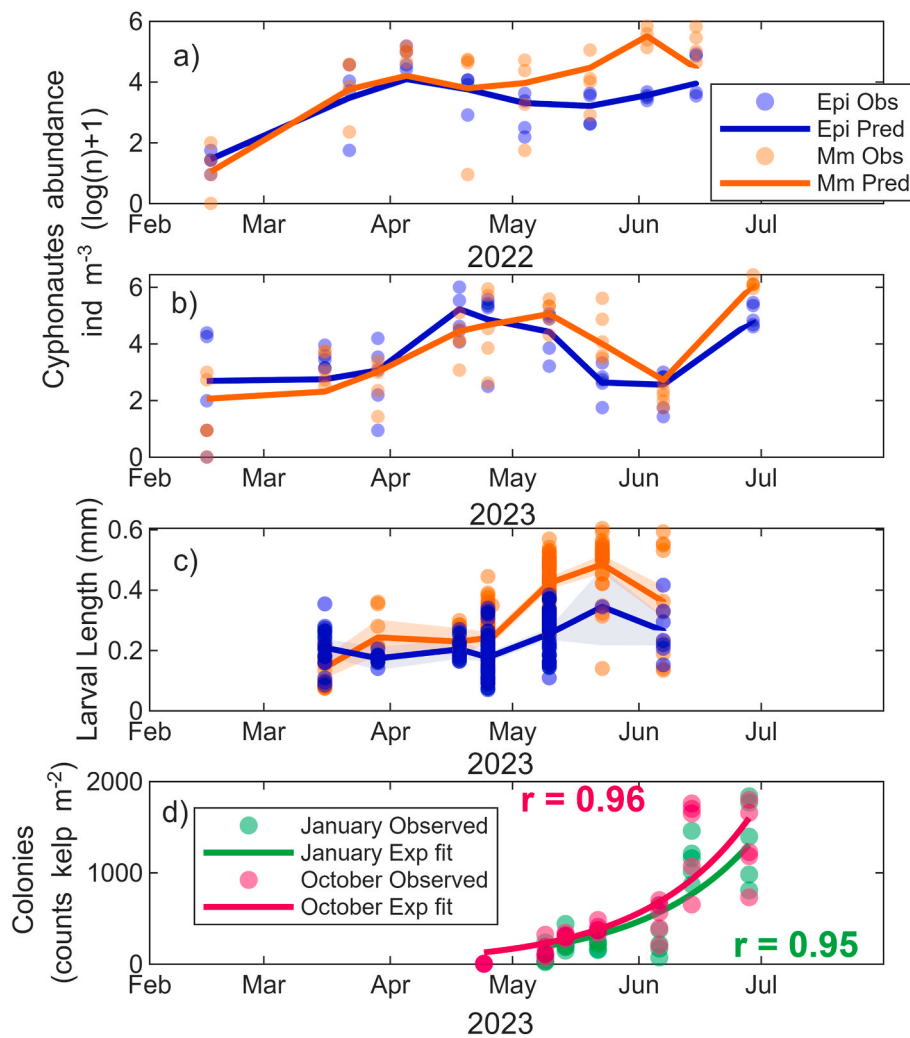


Fig. 6. Observed and predicted seasonal trends in larval abundances of *Electra pilosa* (blue) and *Membranipora membranacea* (orange) from a) 2022 and b) 2023 as well as c) larval size (length) from the water column as well as d) colonization in the kelp in 2023 using generalized linear mixed-effects models (GLMMs). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

April), peaked in mid-April for both years and persisted with high $\text{Chl}_{\text{aframe}}$ concentrations ($\sim 2 \text{ mg m}^{-3}$) throughout the end of June. Likewise, cyphonautes showed two peaks in abundance - one in April/May and a second one in June/July in the studied period, with the first peak occurring 1–2 weeks after the phytoplankton bloom onset. Previous studies reported varying results, with high abundance of *M. membranacea* larvae from January to July in southern California (Yoshioka, 1982), while peak cyphonautes abundance occurred later in other regions: June in central Norway and in September in Nova Scotia (Førde et al., 2016; Saunders and Metaxas, 2010).

High abundance of cyphonautes have usually been found in shallow, warm and low-salinity surface layers, when a strong pycnocline was present, indicating potential onshore transport of the larvae during wind-driven downwelling events (Saunders and Metaxas, 2010). In this study, relatively low salinity (~ 32) was a strong predictor for high abundance of cyphonautes in the water. However, other studies showed the presence of the cyphonautes was most abundant in the upper 5–10 m of the water column during the cooler winter months and most scarce in high temperature, surface waters during summer in California (Yoshioka, 1982). Variability in sampling timing, temperature and frequency could explain the discrepancy in these studies, as well as small-scale patchiness, typical in zooplankton communities in coastal regions, including kelp beds (Fragoso et al., 2022; Maldonado-Aguilar et al., 2023).

While most bryozoan larvae are lecithotrophic (non-feeding and yolk-dependent), cyphonautes larvae of *M. membranacea* and *E. pilosa* are obligatory planktotrophs (feed on plankton) with a functional digestive tract (Woollacott and Zimmer, 2013). This means that these larvae rely on feeding on plankton and other particles and continue to develop in the water, where they can have a long pelagic phase, lasting days to weeks, and potentially as long as two months (Woollacott and Zimmer, 2013). Similarly to other studies found in this region (Fragoso et al., 2021, 2024), $\text{Chl}_{\text{aframe}}$ concentration was relatively high ($> 2 \text{ mg m}^{-3}$) from April until late June and could reflect a source of food that can keep cyphonautes larvae in the water for longer periods. However, our data did not show strong quantitative relationships between phytoplankton size classes and larval abundance, suggesting that phytoplankton size structure alone may not capture the aspects of food quality that influence cyphonautes growth and retention in the water column. While starvation can reduce internal structures of the organism needed for competence and metamorphosis (pyriform organ, internal sac), their shell size and ciliary bands for swimming and feeding are unaffected, allowing them to maintain mobility and survive (Strathmann et al., 2008). Enhanced stratification and northward flow of the NCC, typically increased from spring to late summer, can potentially have a role in the widespread dispersal of the larvae over large areas, allowing them to colonize more diverse habitats and expand their geographical area before reaching metamorphic competence.

4.2. Environmental effects on larval competence, settlement and colony growth

To be able to settle on a kelp lamina, cyphonautes need to reach a certain size for competence before metamorphosis occurs (750 μm in length for *M. membranacea* and 400 μm for *E. pilosa* (Ryland and Stebbing, 1971). In this study, *E. pilosa* larvae from plankton nets were relatively small (0.24 mm basal edge, 0.27 mm height) compared to values from the literature (0.39 mm across the base and 0.36 mm in height) (Atkins, 1955), and are presumably in the pre-competent stage (<440 μm , according to (Saunders and Metaxas, 2010). *M. membranacea* larvae were also smaller (basal edge of 0.62 mm) compared to that found in the literature (0.75–0.8 mm basal edge), but size generally increased by late May/early June, reflecting later stages of development before settling (Atkins, 1955; Ryland and Stebbing, 1971). The variability in size of *M. membranacea* larvae found in our study suggested that a second spawning event possibly occurred in June, likely from colonies already established at the kelp farm or from surrounding wild kelp. Fast competence and presence of multiple cohorts of *M. membranacea* larvae have been suggested as successful traits for settlement in ephemeral habitats, contributing to the invasive potential of this species along the east coast of the US (Saunders and Metaxas, 2010). Other factors may also explain the size patterns observed in *M. membranacea*, highlighting the need for future work that takes further consideration into high-resolution sampling, including depth-stratified sampling, hydrodynamic modelling and larval cohort structure.

Temperature variability is known to affect the growth and development rates of larvae of many marine animals, directly influencing their dispersal distance and survival (O'Connor et al., 2007). In this study, high temperature, low salinity and low Chl_a concentrations were found to be predictors of *M. membranacea* size, while for *E. pilosa*, no significant effect of the environmental conditions on larval size was found. The fact that cyphonautes were found in high abundance in the water but did not settle on the seaweed earlier in the season (early April) suggested that the temperature was not high enough for the larvae to reach competence and, thus, to settle. In this study, the first bryozoan settlement was detected on April 25 at 6.6 °C, with peaks in larval settlement and pre-ancestrulae counts following sharp temperature increases: from 7 °C to ~9 °C, and later from 9.5 °C to 11.5 °C. In other regions, the timing of bryozoan colony appearance has been reported to occur when temperatures were 7.5 °C in early May (Njåstad, 2018) and 10 °C in June in Central Norway (Førde et al., 2016), 13 °C in June in southwest England, UK (Corrigan et al., 2023), while colonies appeared at 6–8 °C in mid-July in Tromsø (Matsson et al., 2019). A recent systematic review showed that bryozoan presence on farmed kelp was associated with temperatures below 11 °C, and because most documented cases come from Norway with far fewer reports further south, bryozoans may be less problematic in more temperate latitudes (Delannoy et al., 2025). The influence of temperature on cyphonautes settling, recruitment and colony growth has been consistently reported in the literature in subarctic regions (Saunders and Metaxas, 2007, 2009, 2010; Scheibling and Gagnon, 2009; Caines and Gagnon, 2012), with models predicting that a 1 °C or 2 °C rise could raise kelp coverage by a factor of 9 and 62, respectively (Saunders et al., 2010). In St. Margarets Bay, Nova Scotia, for example, *M. membranacea* settler abundance was strongly associated with growing degree-days (GDD)—a measure of thermal history—and was 10 × higher following a significantly warmer winter (Saunders and Metaxas, 2007).

High exposure, likewise, could influence recruitment, because strong currents could hinder cyphonautes settlement on the kelp. Intermediate wave exposure in Nova Scotia has been suggested to correlate positively with high numbers of *M. membranacea* settlers on kelp lamina (Caines and Gagnon, 2012). In this study, a storm taking place in early June 2023 (high wind speed, up to 20 m s⁻¹) coincided with intense water mixing, low larval abundances in net samples and low larval settlement and pre-ancestrulae abundance in kelp blades. Visch et al. (2020a) found

that kelp biofouling declined with increasing wave exposure, dropping from 10 % at sheltered sites and 6 % at moderately exposed sites to just 3 % at exposed locations. This contrasts with (Matsson et al., 2019), who observed coverage by *M. membranacea* colonies 64 × higher in more exposed, semi-offshore site when compared to an inshore, fjord site. For both studies, the variation in coverage was high between the locations, even if the distance between locations was small: 20 km away for (Visch et al., 2020) and 50 km away for (Matsson et al., 2019).

Membranipora membranacea larvae use their pyriform organ and mucus to crawl along the kelp lamina, typically moving from older to younger tissue, more proximal regions for settlement (Matson et al., 2010). While the mechanisms guiding this site selection remain unclear, our study found that nearly all larvae and pre-ancestrulae settled on the newer, proximal lamina, indicating a true preference rather than gregariousness or space limitation. Similar patterns have been reported previously (Denley et al., 2014; Matsson et al., 2019), although *E. pilosa* settlers were more abundant on slightly older tissue, likely due to competition with *M. membranacea* (Denley et al., 2014). Heavier fouling was observed on older lamina regions, possibly driving larvae to settle on newer tissue where space is available, and growth time is maximized before tip shedding (Matsson et al., 2019). The sharp increase in colony abundance and coverage in this study likely results from both intensified settlement input and enhanced growth of colonies already present. Another explanation for the rapid proliferation is the much higher density of farmed kelps compared to wild forests, where reduced predation from benthic consumers can facilitate the expansion of bryozoan colonies (Corrigan et al., 2024).

The percentage coverage was higher from the October 2022 deployment than from the January 2023 one. Also, the colony abundance was higher for October with a greater number of larger colonies. Larger colonies, which grew exponentially and thus expand in area more rapidly than smaller ones, had higher coverage (Hermansen et al., 2001; Riisgård and Goldson, 1997; Saunders and Metaxas, 2007). Moreover, the October sporophytes were longer, providing more substrate area for the colonies to expand (Matsson et al., 2021). also observed that earlier deployment times (February compared with April and May) gave a higher biomass of *S. latissima*, and higher biofouling coverage.

Bryozoan colonies need an ambient water flow to receive optimal food supply and intake for growth. For example, water velocity is typically higher in shallower water than in deeper water, and increased water velocity in near-surface kelp farms can enhance food supply and growth rates of filter-feeders (Lenihan et al., 1996; Wilson-Ormond et al., 1997) found that low water velocities limited the flow of food particle flux and slowed oyster growth, suggesting that water flow was more influential than food concentrations. On the other hand (Hermansen et al., 2001), observed no difference in colony growth when *E. pilosa* was exposed to increased water flow velocities (observations from 30 cm s⁻¹ up to 150 cm s⁻¹) (Saunders and Metaxas, 2009). reported an interaction between temperature and food availability, where food limitation reduced colony diameter only at higher temperatures, likely due to elevated metabolic demands. This indicates the existence of inter-dependent relationships between environmental variables (temperature, Chl *a*, current velocity, light) and bryozoan feeding, which needs to be better understood.

5. Conclusions

This study highlights the complex interplay of environmental and biological variables driving bryozoan biofouling at a kelp farm located in Central Norway. While *M. membranacea* and *E. pilosa* larvae peaked in abundance during spring, larval size for *M. membranacea* in the water—rather than abundance alone—was more closely linked to colony settlement, emphasizing the role of larval maturity in biofouling dynamics. This suggests that including larval size monitoring in routine farm surveys could substantially improve early detection of periods for potentially high fouling. Larval abundance was high in stratified, low-

salinity waters. Recruitment was greater on kelp deployed earlier in the season, with substantial increases in colony size and coverage observed from May to June. Key drivers of biofouling were food availability following spring phytoplankton blooms, stable stratified waters, and sudden temperature rises. Such factors influenced larval abundance, settlement timing, and colony development of *M. membranacea* and *E. pilosa* on the kelp.

CRedit authorship contribution statement

G.M. Fragoso: Writing – review & editing, Writing – original draft, Supervision, Project administration, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Nalia Hama:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation. **Elisabeth Snijder:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **David Aldridge:** Writing – review & editing, Writing – original draft, Supervision, Funding acquisition, Conceptualization. **Ana R. Borrero-Santiago:** Writing – review & editing, Supervision. **Ole Jacob Broch:** Writing – review & editing, Supervision. **Geir Johnsen:** Writing – review & editing, Supervision, Funding acquisition. **Yngvar Olsen:** Writing – review & editing, Supervision. **Maja Hatlebakk:** Writing – review & editing, Formal analysis, Data curation.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the G.M.F. used ChatGPT, in certain cases, to improve the quality of the text and assist with coding writing in Matlab. After using this tool, G.M.F. reviewed and edited the content as needed and took full responsibility for the content of the published article.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.marenvres.2026.107854>.

Data availability

Data will be made available on request.

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